SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence **COURSE CODE:** CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. *(BL3)*

Assessment Tool Weightage: 30%

QUESTIONS: 2

Duration : 45 Minutes
Total Marks:20

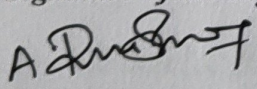
Annexure A

Descriptive Written Test

Code of Conduct:

I A.Rithin Sai (19722600016) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



. smart Home Automation system :-

An intelligent agent is a system that observes its environment process information, and takes suitable actions to achieve its objectives. In a smart home automation system, intelligent agents are used to control lighting, temperature, and security based on user preferences and environmental conditions. Different types of intelligent agents are used to improve the performance of the system.

A simple reflex agent selects actions based on the current situation using predefined rules. For ex, when a motion sensor detects a person entering a room, the system automatically switches on the lights. Similarly, when the room temperature becomes too high, the air conditioner can be switched on. This agents responds quickly to real-time inputs, but it cannot remember past events or learn user preferences.

Iterative Deepening search (IDS) combines the advantages of depth-limited search and breadth-first search. It repeatedly explores the maze with increasing depth limits. Initially, the robot searches paths up to depth one. If the exit is not found, it increases the limit to two and continues the process. This continues until the exit is found. IDS uses less memory because it explores one path at a time making up to depth one. If the exit is not found, it increases the limit to two and continues the process. This continues until the exit is found. IDS uses less memory because it explores one path at a time, making it suitable for robots with limited memory; it is complete and finds the shallowest solution.

The main advantage of UCS is that it finds the least-cost path even when different movements have different costs, but repeated exploration can waste time.

...ubines the
... and brea
... maze
... robot

... in an unknown maze:

A goal based agent selects actions according to specific goals. In a smart home, its goals may include maintaining a comfortable temperature, reducing electricity consumption, and protecting the home from unauthorised entry. For ex: when nobody is at home, it can turn off unnecessary lights and electrical appliances while keeping the security system active. It evaluates possible actions and choose those that help achieve the desired result.

A learning agent can also be included to improve the system over time. it observes user behaviour and learns preferences, such as preferred room temperature lighting brightness and usual schedules. Based on this experience. it can automatically adjust settings to provide a personalised environment.

The best solution is a hybrid intelligent agent that combines the advantages of all these agents. The simple reflex components achieves specifics objectives and the utility-based.